



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT(25CS1302E)

2026-27, Term-IV

Project Problem Statement Submission Form

Sec No-4	Team No-3	Project Title-Smart Expense Splitter and Group Budget Management System
Team Lead Roll Number		2520030412
Team Lead Name		Challa Amrutha
Team Member Roll Number		2520030270
Team Member Name		Dhurga Meenakshi Yenduri
Team Member Roll Number		2520030475
Team Member Name		Earla Ashwith Reddy

1. Abstract

(Summarize the project idea, problem background, and proposed solution. Not more than 250 words.)

The Smart Expense Splitter and Group Budget Management System is designed to simplify shared expense management through intelligent splitting, budgeting, and financial insights. Unlike traditional expense-splitting systems, it provides predictive budget alerts, spending-pattern analysis, what-if budget simulation, and optimized debt settlement. Users can create groups, record expenses, set flexible sharing rules, and track individual contributions. The system analyzes spending trends to predict budget overruns and identify unusual expenses. It also suggests efficient settlements to reduce unnecessary transactions between members. The proposed system aims to make group financial management **smarter, faster, transparent, and more proactive

2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	Assigned Roll Number(s)	Signature
User, Group & Expense Management	2520030412	
Budget, Analytics & Smart Alerts	2520030270	
Settlement & Backend Services	2520030475	

3. Module Details

(Explain the responsibility of each student concerning their assigned module.)

Challa Amrutha – User, Group & Expense Management

Responsible for developing the user and group management module, including user registration, authentication, group creation, member management, and expense recording. This module will also handle different expense-sharing rules such as equal, percentage-based, and custom splitting. The student will work on database design and APIs required for managing users, groups, expenses, and contributions.



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Dhurga Meenakshi Yenduri -Budget, Analytics & Smart Alerts

Responsible for the budgeting and intelligent financial analysis module. This includes setting group budgets, tracking spending against budgets, analyzing spending patterns, predicting possible budget overruns, and identifying unusual expenses. The student will also implement what-if budget simulation and generate useful alerts and financial insights for users.

Earla Ashwith Reddy – Settlement & Backend Services

Responsible for the debt settlement and backend service module. This includes calculating member balances, optimizing settlements to reduce unnecessary transactions, and developing the required backend APIs and services. The student will also contribute to microservice integration, database connectivity, testing, and deployment-related components.

4. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** FastAPI, Node.js, Express.js, Spring Boot
- **Databases:** PostgreSQL, MongoDB
- **Vector Database:** pgvector
- **API & Testing:** REST API, OpenAPI/Swagger, Postman, Pytest
- **Authentication & Security:** JWT, OAuth2, RBAC
- **Deployment:** Docker, Docker Compose
- **Monitoring:** Prometheus, Grafana, OpenTelemetry
- **Development Tools:** Visual Studio Code, Git, GitHub
- **Programming Languages:** Python, Java, JavaScript/Node.js

Guide Name and Signature (remarks if any) _____